Optimal control of number squeezing in trapped Bose-Einstein condensates

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(Received 31 August 2009; published 24 November 2009)

We theoretically analyze atom interferometry based on trapped ultracold atoms and employ optimal control theory in order to optimize number squeezing and condensate trapping. In our simulations, we consider a setup where the confinement potential is transformed from a single to a double well, which allows us to split the condensate. To avoid in the ensuing phase-accumulation stage of the interferometer dephasing due to the nonlinear atom-atom interactions, the atom-number fluctuations between the two wells should be sufficiently low. We show that low number fluctuations (high number squeezing) can be obtained by optimized splitting protocols. Two types of solutions are found: in the Josephson regime we find an oscillatory tunnel control and a parametric amplification of number squeezing, while in the Fock regime squeezing is obtained solely due to the nonlinear coupling, which is transformed to number squeezing by peaked tunnel pulses. We study splitting and squeezing within the frameworks of a generic two-mode model, which allows us to study the basic physical mechanisms, and the multiconfigurational time-dependent Hartree for bosons method, which allows for a microscopic modeling of the splitting dynamics in realistic experiments. Both models give similar results, thus highlighting the general nature of these two solution schemes. We finally analyze our results in the context of atom interferometry.

DOI: 10.1103/PhysRevA.80.053625 PACS number(s): 03.75.−b, 03.75.Mn, 37.25.+k, 03.75.Gg

I. INTRODUCTION

Squeezed states offer the possibility of measurements below shot noise [1,2]. In matter wave interferometry [3] with Bose-Einstein condensates (BECs), number squeezing in the split BEC helps us to slow down the phase diffusion resulting from atom-atom interactions [4] and thus the possible coherence time, and with it the sensitivity of the atom interferometer, is increased.

Splitting a trapped BEC is achieved by transforming the confinement potential from a single to a double well [5,6]. Close to the splitting point, where the condensate breaks apart into two spatially separated parts, the competition between tunneling and nonlinear interactions leads to a decrease of number fluctuations and squeezing.

In this paper we will discuss the concepts to optimize squeezing and coherence in trapped BECs in the context of atom chips [7–9] which provide a versatile tool for precise manipulation of trapped BECs and allow robust implementations of the splitting procedure and interference experiments [6,10]. A key ingredient in our considerations is the beam splitter for the trapped or guided atoms [11–13]. As an example we will consider the rf-induced beam splitter [14,15] as employed in the atom chip experiments. However, our considerations can be easily applied to any other double well splitting and trapped atom interferometry setup [5,16,17,18].

Squeezing in the context of BEC splitting has been frequently discussed by assuming an approximately exponential decrease of the tunnel coupling close to the splitting point [19–22]. In experiment, number squeezing has been achieved in an optical trap [23], in optical lattices [24–26], and more recently through rf-splitting [10,27]. In Ref. [28] we have recently shown that exponential splitting is by far not optimal, and have constructed alternative squeezing protocols allowing for high number squeezing. These solutions are obtained within the framework of optimal control theory (OCT) [29,30]. In this paper we give details on our work about optimal number squeezing.

In our simulations, we go beyond the simple two-mode model and consider additionally the spatial dynamics in order to have a realistic description of the splitting dynamics and to evaluate the importance of condensate excitations. Our theoretical approach is based on the multiconfigurational time-dependent Hartree for bosons (MCTDHB) method [31], which allows us to describe both atom-number and spatial dynamics on a self-consistent footing. This method allows us to use as a control parameter the splitting distance of the trap used in experiments. The generic two-mode model, whose control is through the tunnel coupling which is determined indirectly through the confinement potential, is used to obtain a detailed understanding of the underlying control strategies.

We find that these yield rather complex splitting protocols compared to our earlier work [28]. These strategies can be understood similarly to the control schemes in Refs. [34,35] which rely on a constant optimal value of the tunnel coupling. We compare the OCT results to a two-parameter optimization where the control consists of an initial exponential splitting followed by a constant value of the tunnel coupling. While within the generic two-mode model the results almost coincide, this simple strategy is not generally applicable within a realistic model for two main reasons: first, for slow splitting times on a time scale determined by the trapping potential the condensate oscillates, and the split condensates end up in an excited state; second, after number squeezing the condensates have to be further separated to turn off any tunnel coupling, which for realistic traps can only be achieved with nonadiabatic protocols. To avoid these problems, we

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